

SAT UZBEKISTAN

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Module-1

1

Mark for Review



For a particular machine that produces beads, 29 out of every 100 beads it produces have a defect. A bead produced by the machine will be selected at random. What is the probability of selecting a bead that has a defect?

(A) $\frac{1}{2,900}$

(B) $\frac{1}{29}$

(C) $\frac{29}{100}$

(D) $\frac{29}{10}$

2

Mark for Review



$$s = 40 + 3t$$

The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

(A) 40

(B) 43

(C) 45

(D) 55

3

Mark for Review



If $4x = 3$, what is the value of $24x$?

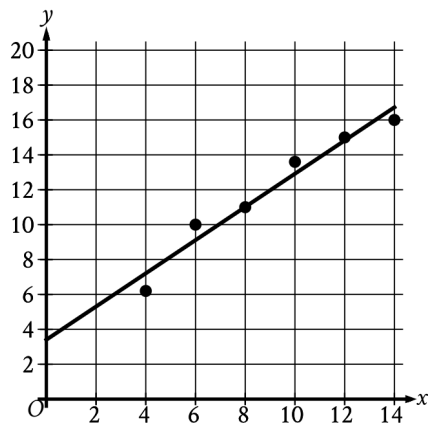
(A) $\frac{9}{2}$

(B) 6

(C) 18

(D) 72

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

(A) $y = x + 3.4$

(B) $y = x - 3.4$

(C) $y = -x + 3.4$

(D) $y = -x - 3.4$

Of 300,000 paper clips, 234,000 are size large. What percentage of the paper clips are size large?

(A) 22%

(B) 33%

(C) 66%

(D) 78%

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

(A) $y = 18x + 13$

(B) $y = 18x + 18$

(C) $y = -5x + 13$

(D) $y = -5x + 18$

The function f is defined by $f(x) = x^2 + x + 71$. What is the value of $f(2)$?

Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m , q , and z are positive?

(A) $m^4q^{20}z^{-3}$

(B) $m^5q^9z^2$

(C) $m^6q^8z^{-1}$

(D) $m^{20}q^{12}z^{-2}$



Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

(A) 204

(B) 182

(C) 60

(D) 12

If $\frac{x}{y} = 4$ and $\frac{24x}{ny} = 4$, what is the value of n ?



$$w(t) = 300 - 4t$$

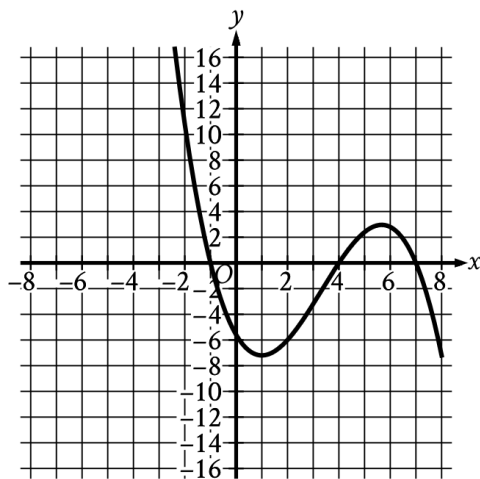
The function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. According to the model, what is the predicted volume, in milliliters, draining from the container each second?

(A) 300

(B) 296

(C) 75

(D) 4



The graph of $y = f(x)$ is shown, where the function f is defined by $f(x) = ax^3 + bx^2 + cx + d$ and a , b , c , and d are constants. For how many values of x does $f(x) = 0$?

- (A) One
- (B) Two
- (C) Three
- (D) Four

$$f(x) = 3,000(0.75)^x$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 3,000 in this context?

- (A) The estimated percent decrease in the population for this species and area every 8 years after 2008
- (B) The estimated percent decrease in the population for this species and area each year after 2008
- (C) The estimated population for this species and area 8 years after 2008
- (D) The estimated initial population for this species and area in 2008

14

 Mark for Review

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y . For what value of x does the value of y reach its minimum?

15

 Mark for Review

Square A has side lengths that are 166 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

16

 Mark for Review

$$2a + 8b = 198$$

$$2a + 4b = 98$$

The solution to the given system of equations is (a, b) . What is the value of b ?

17

 Mark for Review

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

(A) $(0, 14)$

(B) $(0, 2)$

(C) $(0, 22)$

(D) $(0, -8)$

18

 Mark for Review

Two variables, x and y , are related such that for each increase of 1 in the value of x , the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

(A) $y = 4(x)^{200}$

(B) $y = 4(200)^x$

(C) $y = 200(x)^4$

(D) $y = 200(4)^x$

What is the value of $\sin 42\pi$?

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{\sqrt{2}}{2}$

(D) 1

$$4x - 9y = 9y + 5$$

$$hy = 2 + 4x$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h ?

(A) -9

(B) 0

(C) 9

(D) 18

$$x^2 - 2x - 9 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

(A) 8

(B) 10

(C) 20

(D) 40

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

In the given equation, c is a positive constant. Which of the following is one of the solutions to the given equation?

(A) $-c$

(B) $-c^2 - 39^2$

(C) $-\sqrt{39^2 - c^2}$

(D) $-\sqrt{c^2 + 39^2}$

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Module-2

1

Mark for Review



An airplane descends from an altitude of 9,500 feet to 5,000 feet at a constant rate of 400 feet per minute. What type of function best models the relationship between the descending airplane's altitude and time?

(A) Decreasing exponential

(B) Decreasing linear

(C) Increasing exponential

(D) Increasing linear

2

Mark for Review



Line k is defined by $y = \frac{17}{7}x + 4$. Line j is parallel to line k in the xy -plane. What is the slope of line j ?

(A) $\frac{7}{17}$

(B) $\frac{17}{7}$

(C) 4

(D) 17

Caleb used juice to make popsicles. The function $f(x) = -5x + 30$ approximates the volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. Which statement is the best interpretation of the y -intercept of the graph of $y = f(x)$ in the xy -plane in this context?

- (A) Caleb used approximately 5 fluid ounces of juice for each popsicle.
- (B) Caleb had approximately 5 fluid ounces of juice when he began to make the popsicles.
- (C) Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.
- (D) Caleb used approximately 30 fluid ounces of juice for each popsicle.

A physics class is planning an experiment about a toy rocket. The equation $y = -16(x - 5.6)^2 + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

- (A) This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.
- (B) This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.
- (C) This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.
- (D) This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

The function f is defined by $f(x) = 4x + k(x - 1)$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

(A) 18°

(B) 72°

(C) 90°

(D) 162°

$$y \leq x + 7$$

$$y \geq -2x - 1$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

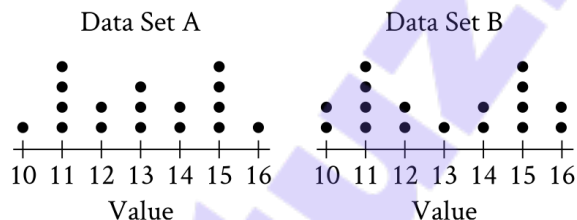
(A) $(-14, 0)$

(B) $(0, -14)$

(C) $(0, 14)$

(D) $(14, 0)$

The dot plots represent the distributions of values in data sets A and B.



Which of the following statements must be true?

- I. The median of data set A is equal to the median of data set B.
- II. The standard deviation of data set A is equal to the standard deviation of data set B.

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II



A scientist initially measures 12,000 bacteria in a growth medium. 4 hours later, the scientist measures 24,000 bacteria. Assuming exponential growth, the formula $P = C(2)^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r ?

(A) $\frac{1}{12,000}$

(B) $\frac{1}{4}$

(C) 4

(D) 12,000

A cube has a volume of 474,552 cubic units. What is the surface area, in square units, of the cube?

$$2(8x) + 4(7y) = 12$$

$$-2(8x) + 4(7y) = 12$$

The solution to the given system of equations is (x, y) . What is the value of $8x + 7y$?



A certain town has an area of 4.36 square miles. What is the area, in square yards, of this town? (1 mile = 1,760 yards)

(A) 404

(B) 7,674

(C) 710,459

(D) 13,505,536

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $(4, 5)$, $(4, 7)$, and $(6, 5)$, respectively. Triangle LMN has vertices L , M , and N at $(4, 5)$, $(4, 7 + k)$, and $(6 + k, 5)$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

(A) $(90 - (t - k))^\circ$

(B) $(90 - (t + k))^\circ$

(C) $(90 - t)^\circ$

(D) $(90 + k)^\circ$

A small business owner budgets \$2,200 to purchase candles. The owner must purchase a minimum of 200 candles to maintain the discounted pricing. If the owner pays \$4.90 per candle to purchase small candles and \$11.60 per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

(A) 20

(B) $\frac{20\sqrt{2}}{2}$

(C) $20\sqrt{2}$

(D) 40

Which of the following equations represents a circle in the xy -plane that intersects the y -axis at exactly one point?

(A) $(x - 8)^2 + (y - 8)^2 = 16$

(B) $(x - 8)^2 + (y - 4)^2 = 16$

(C) $(x - 4)^2 + (y - 9)^2 = 16$

(D) $x^2 + (y - 9)^2 = 16$

$$y = 6x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

(A) $-6x + y = 18$

(B) $-6x + y = 22$

(C) $-12x + y = 36$

(D) $-12x + y = 18$

Which expression is equivalent to $\frac{y+12}{x-8} + \frac{y(x-8)}{x^2y-8xy}$?

(A) $\frac{xy+y+4}{x^3y-16x^2y+64xy}$

(B) $\frac{xy+9y+12}{x^2y-8xy+x-8}$

(C) $\frac{xy^2+13xy-8y}{x^2y-8xy}$

(D) $\frac{xy^2+13xy-8y}{x^3y-16x^2y+64xy}$



$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

(A) $p + r + s$

(B) $20(p + r + s)$

(C) $\frac{prs}{pr+ps+rs}$

(D) $\frac{prs}{20p+20r+20s}$

$$x(kx - 56) = -16$$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?



$$\begin{aligned} 2x + 3y &= 7 \\ 10x + 15y &= 35 \end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

(A) $\left(\frac{r}{5} + 7, -\frac{r}{5} + 35\right)$

(B) $\left(-\frac{3r}{2} + \frac{7}{2}, r\right)$

(C) $\left(r, \frac{2r}{3} + \frac{7}{3}\right)$

(D) $\left(r, -\frac{3r}{2} + \frac{7}{2}\right)$



A window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is \$400. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

(A) $f(x) = 60x + 100$

(B) $f(x) = 60x + 220$

(C) $f(x) = 80x$

(D) $f(x) = 80x + 220$